2018 CATHOLIC HIGH SCHOOL CALCULUS A INDIVIDUAL

1. Find $\lim_{x \to \pi/2} \left(\frac{\cos^2(x)}{1 - \sin x} \right)$

2. Find $\lim_{x \to 2} \left(\frac{e^x - e^2}{x - 2} \right)$

3. Find the equation of the line tangent to $y = \ln x$ when $x = e$ (answer in slope intercept form)

4. Find the coordinates of all points on $y = x^3 - x$ that have tangent lines parallel to $y = 2x$

5. If $j(x) = \frac{f(x)}{x}$ and $f(x)$ has values given in the table below find $j'(3)$

x	$f(x)$	$f'(x)$
3	2	-2

6. Find the value of the derivative for $x = \tan y$ at $\left(1, \frac{\pi}{4}\right)$

7. Find the slope of the normal to $x^3 + 2xy = 5$ at $(1,2)$

8. If $y = \ln(5x)$, find $y'''(3)$

9. The tangent line to $f(x)$ at $(2,7)$ also passes through $(4,-9)$.
Find $f(2)+f'(2)$

10. If $f(x)=x-2\sin x$ on $[0,2\pi]$, give the x coordinate of the relative maximum point

11. On what open interval(s) is $f(x)=x^3-6x^2+15$ decreasing?

12. Find the maximum value of $f(\theta)=\theta+\cos\theta$ on $[0,\pi]$

13. If the position of a particle moving along a horizontal number line is given by $x=2t^3+3t^2-36t+40$ (x in cm, t in seconds), find the acceleration (in cm/sec^2) when the velocity is 0.

14. The area of a rectangle is increasing at 14 cm^2/sec and the length of the rectangle is decreasing at 2 cm/sec. How fast is the width changing when the length is 12 cm and the area is 60 cm^2? (include units)

15. Find c for which $y=x^2+c$ is tangent to the line $y=x$.

Catholic High Solutions

1 - 2

2 - e^2

3 - $y = \frac{1}{e}x$

4 - $(1, 0), (-1, 0)$

5 - $\frac{-8}{9}$

6 - $\frac{1}{2}$

7 - $\frac{2}{7}$

8 - $\frac{2}{27}$

9 - -1

10 - $\frac{5\pi}{3}$

11 - $(0, 4)$

12 - $\pi - 1$

13 - 30 cm/sec^2

14 - 2 cm/sec

15 - $c = \frac{1}{4}$